12.0 INTERRUPTS

The ENC28J60 has multiple interrupt sources and an interrupt output pin to signal the occurrence of events to the host controller. The interrupt pin is designed for use by a host controller that is capable of detecting falling edges.

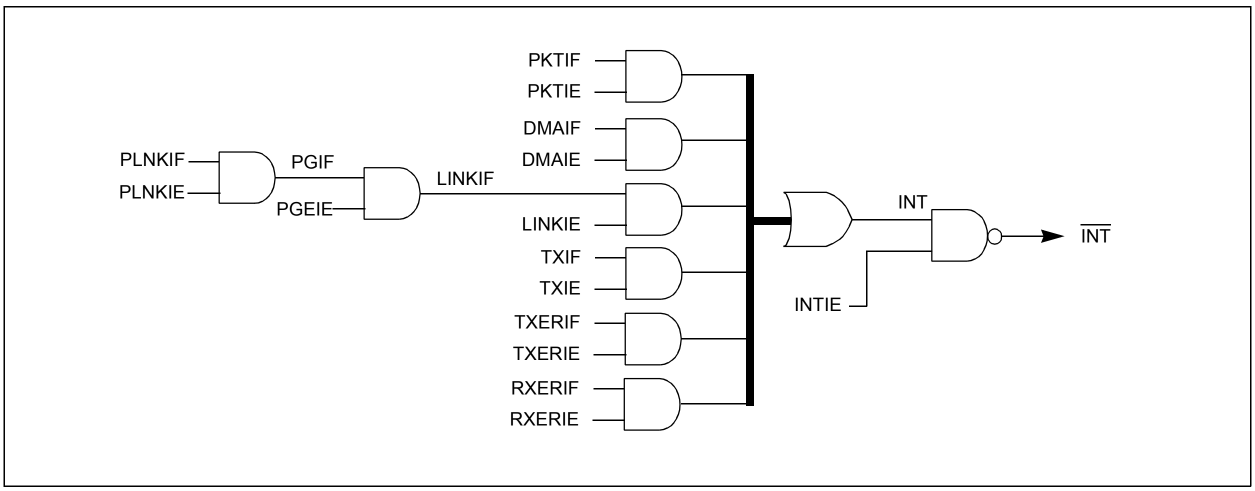
Interrupts are managed with two registers. The EIE register contains the individual interrupt enable bits for each interrupt source, while the EIR register contains the corresponding interrupt flag bits. When an interrupt occurs, the interrupt flag is set. If the interrupt is enabled in the EIE register and the INTIE global interrupt enable bit is set, the $\overline{\text{INT}}$ pin will be driven low (see Figure 12-1).

Note:	Except for the LINKIF interrupt flag, interrupt flag bits are set when an interrupt condition occurs regardless of the state of its corresponding enable bit or the associated global enable bit. User software should ensure the appropriate interrupt flag bits are clear prior to enabling an interrupt. This feature allows for software polling.

When an enabled interrupt occurs, the interrupt pin will remain low until all flags which are causing the interrupt are cleared or masked off (enable bit is cleared) by the host controller. If more than one interrupt source is enabled, the host controller must poll each flag in the EIR register to determine the source(s) of the interrupt. It is recommended that the Bit Field Clear (BFC) SPI command be used to reset the flag bits in the EIR register rather than the normal Write Control Register (WCR) command. This is necessary to prevent unintentionally altering a flag that changes during the write command. The BFC and WCR commands are discussed in detail in **Section 4.0 "Serial Peripheral Interface (SPI)"**.

After an interrupt occurs, the host controller should clear the global enable bit for the interrupt pin before servicing the interrupt. Clearing the enable bit will cause the interrupt pin to return to the non-asserted state (high). Doing so will prevent the host controller from missing a falling edge should another interrupt occur while the immediate interrupt is being serviced. After the interrupt has been serviced, the global enable bit may be restored. If an interrupt event occurred while the previous interrupt was being processed, the act of resetting the global enable bit will cause a new falling edge on the interrupt pin to occur.

FIGURE 12-1: ENC28J60 INTERRUPT LOGIC



12.1 INT Interrupt Enable (INTIE)

The INT Interrupt Enable bit (INTIE) is a global enable bit which allows the following interrupts to drive the $\overline{\text{INT}}$ pin:

- Receive Error Interrupt (RXERIF)
- Transmit Error Interrupt (TXERIF)
- Transmit Interrupt (TXIF)
- Link Change Interrupt (LINKIF)
- DMA Interrupt (DMAIF)
- Receive Packet Pending Interrupt (PKTIF)

When any of the above interrupts are enabled and generated, the virtual bit, INT in the ESTAT register (Register 12-1), will be set to '1'. If EIE.INTIE is '1', the $\overline{\text{INT}}$ pin will be driven low.

12.1.1 INT INTERRUPT REGISTERS

The registers associated with the INT interrupts are shown in Register 12-2, Register 12-3, Register 12-4 and Register 12-5.

REGISTER 12-1: ESTAT: ETHERNET STATUS REGISTER

R-0	R/C-0	R-0	R/C-0	U-0	R-0	R/C-0	R/W-0
INT	BUFER	r	LATECOL	—	RXBUSY	TXABRT	CLKRDY(1)
bit 7							bit 0

Legend:			
R = Readable bit	C = Clearable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

bit 7 **INT:** INT Interrupt Flag bit
1 = INT interrupt is pending
0 = No INT interrupt is pending

bit 6 **BUFER:** Ethernet Buffer Error Status bit
1 = An Ethernet read or write has generated a buffer error (overrun or underrun)
0 = No buffer error has occurred

bit 5 **Reserved:** Read as '0'

bit 4 **LATECOL:** Late Collision Error bit
1 = A collision occurred after 64 bytes had been transmitted
0 = No collisions after 64 bytes have occurred

bit 3 **Unimplemented:** Read as '0'

bit 2 **RXBUSY:** Receive Busy bit
1 = Receive logic is receiving a data packet
0 = Receive logic is Idle

bit 1 **TXABRT:** Transmit Abort Error bit
1 = The transmit request was aborted
0 = No transmit abort error

bit 0 **CLKRDY:** Clock Ready bit(1)
1 = OST has expired; PHY is ready
0 = OST is still counting; PHY is not ready

Note 1: CLKRDY resets to '0' on Power-on Reset but is unaffected on all other Resets.

REGISTER 12-2: EIE: ETHERNET INTERRUPT ENABLE REGISTER

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
INTIE	PKTIE	DMAIE	LINKIE	TXIE	r	TXERIE	RXERIE
bit 7							bit 0

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as ‘0’	
-n = Value at POR	‘1’ = Bit is set	‘0’ = Bit is cleared	x = Bit is unknown

bit 7 **INTIE:** Global INT Interrupt Enable bit
1 = Allow interrupt events to drive the $\overline{\text{INT}}$ pin
0 = Disable all $\overline{\text{INT}}$ pin activity (pin is continuously driven high)

bit 6 **PKTIE:** Receive Packet Pending Interrupt Enable bit
1 = Enable receive packet pending interrupt
0 = Disable receive packet pending interrupt

bit 5 **DMAIE:** DMA Interrupt Enable bit
1 = Enable DMA interrupt
0 = Disable DMA interrupt

bit 4 **LINKIE:** Link Status Change Interrupt Enable bit
1 = Enable link change interrupt from the PHY
0 = Disable link change interrupt

bit 3 **TXIE:** Transmit Enable bit
1 = Enable transmit interrupt
0 = Disable transmit interrupt

bit 2 **Reserved:** Maintain as ‘0’

bit 1 **TXERIE:** Transmit Error Interrupt Enable bit
1 = Enable transmit error interrupt
0 = Disable transmit error interrupt

bit 0 **RXERIE:** Receive Error Interrupt Enable bit
1 = Enable receive error interrupt
0 = Disable receive error interrupt

REGISTER 12-3: EIR: ETHERNET INTERRUPT REQUEST (FLAG) REGISTER

U-0	R-0	R/C-0	R-0	R/C-0	R-0	R/C-0	R/C-0
—	PKTIF	DMAIF	LINKIF	TXIF	r	TXERIF	RXERIF
bit 7							bit 0

Legend:			
R = Readable bit	C = Clearable bit	U = Unimplemented bit, read as ‘0’	
-n = Value at POR	‘1’ = Bit is set	‘0’ = Bit is cleared	x = Bit is unknown

bit 7 **Unimplemented:** Read as ‘0’

bit 6 **PKTIF:** Receive Packet Pending Interrupt Flag bit
1 = Receive buffer contains one or more unprocessed packets; cleared when PKTDEC is set
0 = Receive buffer is empty

bit 5 **DMAIF:** DMA Interrupt Flag bit
1 = DMA copy or checksum calculation has completed
0 = No DMA interrupt is pending

bit 4 **LINKIF:** Link Change Interrupt Flag bit
1 = PHY reports that the link status has changed; read PHIR register to clear
0 = Link status has not changed

bit 3 **TXIF:** Transmit Interrupt Flag bit
1 = Transmit request has ended
0 = No transmit interrupt is pending

bit 2 **Reserved:** Maintain as ‘0’

bit 1 **TXERIF:** Transmit Error Interrupt Flag bit
1 = A transmit error has occurred
0 = No transmit error has occurred

bit 0 **RXERIF:** Receive Error Interrupt Flag bit
1 = A packet was aborted because there is insufficient buffer space or the packet count is 255
0 = No receive error interrupt is pending

REGISTER 12-4: PHIE: PHY INTERRUPT ENABLE REGISTER

R-0	R-0	R-0	R-0	R-0	R-0	R-0	R-0
r	r	r	r	r	r	r	r
bit 15							bit 8

R-0	R-0	R/W-0	R/W-0	R-0	R-0	R/W-0	R/W-0
r	r	r	PLNKIE	r	r	PGEIE	r
bit 7							bit 0

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as ‘0’	
-n = Value at POR	‘1’ = Bit is set	‘0’ = Bit is cleared	x = Bit is unknown

bit 15-6 **Reserved:** Write as ‘0’, ignore on read

bit 5 **Reserved:** Maintain as ‘0’

bit 4 **PLNKIE:** PHY Link Change Interrupt Enable bit
1 = PHY link change interrupt is enabled
0 = PHY link change interrupt is disabled

bit 3-2 **Reserved:** Write as ‘0’, ignore on read

bit 1 **PGEIE:** PHY Global Interrupt Enable bit
1 = PHY interrupts are enabled
0 = PHY interrupts are disabled

bit 0 **Reserved:** Maintain as ‘0’

REGISTER 12-5: PHIR: PHY INTERRUPT REQUEST (FLAG) REGISTER

R-x	R-x	R-x	R-x	R-x	R-x	R-x	R-x
r	r	r	r	r	r	r	r
bit 15							bit 8

R-x	R-x	R-0	R/SC-0	R-0	R/SC-0	R-x	R-0
r	r	r	PLNKIF	r	PGIF	r	r
bit 7							bit 0

Legend:			
R = Readable bit	SC = Self-clearing bit	U = Unimplemented bit, read as ‘0’	
-n = Value at POR	‘1’ = Bit is set	‘0’ = Bit is cleared	x = Bit is unknown

bit 15-6 **Reserved:** Do not modify

bit 5 **Reserved:** Read as ‘0’

bit 4 **PLNKIF:** PHY Link Change Interrupt Flag bit
1 = PHY link status has changed since PHIR was last read; resets to ‘0’ when read
0 = PHY link status has not changed since PHIR was last read

bit 3 **Reserved:** Read as ‘0’

bit 2 **PGIF:** PHY Global Interrupt Flag bit
1 = One or more enabled PHY interrupts have occurred since PHIR was last read; resets to ‘0’ when read
0 = No PHY interrupts have occurred

bit 1 **Reserved:** Do not modify

bit 0 **Reserved:** Read as ‘0’

12.1.2 RECEIVE ERROR INTERRUPT FLAG (RXERIF)

The Receive Error Interrupt Flag (RXERIF) is used to indicate a receive buffer overflow condition. Alternately, this interrupt may indicate that too many packets are in the receive buffer and more cannot be stored without overflowing the EPKTCNT register.

When a packet is being received and the receive buffer runs completely out of space, or EPKTCNT is 255 and cannot be incremented, the packet being received will be aborted (permanently lost) and the EIR.RXERIF bit will be set to '1'. Once set, RXERIF can only be cleared by the host controller or by a Reset condition. If the receive error interrupt and INT interrupt are enabled (EIE.RXERIE = 1 and EIE.INTIE = 1), an interrupt is generated by driving the $\overline{\text{INT}}$ pin low. If the receive error interrupt is not enabled (EIE.RXERIE = 0 or EIE.INTIE = 0), the host controller may poll the ENC28J60 for the RXERIF and take appropriate action.

Normally, upon the receive error condition, the host controller would process any packets pending from the receive buffer and then make additional room for future packets by advancing the ERXRDPT registers (low byte first) and decrementing the EPKTCNT register. See **Section 7.2.4 "Freeing Receive Buffer Space"** for more information on processing packets. Once processed, the host controller should use the BFC command to clear the EIR.RXERIF bit.

12.1.3 TRANSMIT ERROR INTERRUPT FLAG (TXERIF)

The Transmit Error Interrupt Flag (TXERIF) is used to indicate that a transmit abort has occurred. An abort can occur because of any of the following:

1. Excessive collisions occurred as defined by the Retransmission Maximum (RETMAX) bits in the MACLCON1 register.
2. A late collision occurred as defined by the Collision Window (COLWIN) bits in the MACLCON2 register.
3. A collision after transmitting 64 bytes occurred (ESTAT.LATECOL set).
4. The transmission was unable to gain an opportunity to transmit the packet because the medium was constantly occupied for too long. The deferral limit (2.4287 ms) was reached and the MACON4.DEFER bit was clear.
5. An attempt to transmit a packet larger than the maximum frame length defined by the MAMXFL registers was made without setting the MACON3.HFRMEN bit or per packet POVERRIDE and PHUGEEN bits.

Upon any of these conditions, the EIR.TXERIF flag is set to '1'. Once set, it can only be cleared by the host controller or by a Reset condition. If the transmit error interrupt is enabled (EIE.TXERIE = 1 and EIE.INTIE = 1), an interrupt is generated by driving the $\overline{\text{INT}}$ pin low for one OSC1 period. If the transmit error interrupt is not enabled (EIE.TXERIE = 0 or EIE.INTIE = 0), the host controller may poll the ENC28J60 for the TXERIF and take appropriate action. Once the interrupt is processed, the host controller should use the BFC command to clear the EIR.TXERIF bit.

After a transmit abort, the TXRTS bit will be cleared, the ESTAT.TXABRT bit will be set and the transmit status vector will be written at ETXND + 1. The MAC will not automatically attempt to retransmit the packet. The host controller may wish to read the transmit status vector and LATECOL bit to determine the cause of the abort. After determining the problem and solution, the host controller should clear the LATECOL (if set) and TXABRT bits so that future aborts can be detected accurately.

In Full-Duplex mode, condition 5 is the only one that should cause this interrupt. Collisions and other problems related to sharing the network are not possible on full-duplex networks. The conditions which cause the transmit error interrupt meet the requirements of the transmit interrupt. As a result, when this interrupt occurs, TXIF will also be simultaneously set.

12.1.4 TRANSMIT INTERRUPT FLAG (TXIF)

The Transmit Interrupt Flag (TXIF) is used to indicate that the requested packet transmission has ended (ECON1.TXRTS has transitioned from '1' to '0'). Upon transmission completion, abort or transmission cancellation by the host controller, the EIR.TXIF flag will be set to '1'. If the host controller did not clear the TXRTS bit and the ESTAT.TXABRT bit is not set, then the packet was successfully transmitted. Once TXIF is set, it can only be cleared by the host controller or by a Reset condition. If the transmit interrupt is enabled (EIE.TXIE = 1 and EIE.INTIE = 1), an interrupt is generated by driving the $\overline{\text{INT}}$ pin low. If the transmit interrupt is not enabled (EIE.TXIE = 0 or EIE.INTIE = 0), the host controller may poll the ENC28J60 for the TXIF bit and take appropriate action. Once processed, the host controller should use the BFC command to clear the EIR.TXIF bit.

12.1.5 LINK CHANGE INTERRUPT FLAG (LINKIF)

The LINKIF indicates that the link status has changed. The actual current link status can be obtained from the PHSTAT1.LLSTAT or PHSTAT2.LSTAT (see Register 3-5 and Register 3-6). Unlike other interrupt sources, the link status change interrupt is created in the integrated PHY module; additional steps must be taken to enable it.

By Reset default, LINKIF is never set for any reason. To receive it, the host controller must set the PHIE.PLNKIE and PGEIE bits. After setting the two PHY interrupt enable bits, the LINKIF bit will then shadow the contents of the PHIR.PGIF bit. The PHY only supports one interrupt, so the PGIF bit will always be the same as the PHIR.PLNKIF bit (when both PHY enable bits are set).

Once LINKIF is set, it can only be cleared by the host controller or by a Reset. If the link change interrupt is enabled (EIE.LINKIE = `1`, EIE.INTIE = `1`, PHIE.PLNKIE = `1` and PHIE.PGEIE = `1`), an interrupt will be generated by driving the $\overline{\text{INT}}$ pin low. If the link change interrupt is not enabled (EIE.LINKIE = `0`, EIE.INTIE = `0`, PHIE.PLNKIE = `0` or PHIE.PGEIE = `0`), the host controller may poll the ENC28J60 for the PHIR.PLNKIF bit and take appropriate action.

The LINKIF bit is read-only. Because reading from PHY registers requires non-negligible time, the host controller may instead set PHIE.PLNKIE and PHIE.PGEIE and then poll the EIR.LINKIF bit. Performing an MII read on the PHIR register will clear the LINKIF, PGIF and PLNKIF bits automatically and allow for future link status change interrupts. See **Section 3.3 “PHY Registers”** for information on accessing the PHY registers.

12.1.6 DMA INTERRUPT FLAG (DMAIF)

The DMA interrupt indicates that the DMA module has completed its memory copy or checksum calculation (ECON1.DMAST has transitioned from ‘`1`’ to ‘`0`’). Additionally, this interrupt will be caused if the host controller cancels a DMA operation by manually clearing the DMAST bit. Once set, DMAIF can only be cleared by the host controller or by a Reset condition. If the DMA interrupt is enabled (EIE.DMAIE = `1` and EIE.INTIE = `1`), an interrupt is generated by driving the $\overline{\text{INT}}$ pin low. If the DMA interrupt is not enabled (EIE.DMAIE = `0` or EIE.INTIE = `0`), the host controller may poll the ENC28J60 for the DMAIF and take appropriate action. Once processed, the host controller should use the BFC command to clear the EIR.DMAIF bit.

12.1.7 RECEIVE PACKET PENDING INTERRUPT FLAG (PKTIF)

The Receive Packet Pending Interrupt Flag (PKTIF) is used to indicate the presence of one or more data packets in the receive buffer and to provide a notification means for the arrival of new packets. When the receive buffer has at least one packet in it, EIR.PKTIF will be set. In other words, this interrupt flag will be set anytime the Ethernet Packet Count register (EPKTCNT) is non-zero. If the receive packet pending interrupt is enabled (EIE.PKTIE = `1` and EIE.INTIE = `1`), an interrupt will be generated by driving the $\overline{\text{INT}}$ pin low whenever a new packet is successfully received and written into the receive buffer. If the receive packet pending interrupt is not enabled (EIE.PKTIE = `0` or EIE.INTIE = `0`), the host controller will not be notified when new packets arrive. However, it may poll the PKTIF bit and take appropriate action.

The PKTIF bit can only be cleared by the host controller or by a Reset condition. In order to clear PKTIF, the EPKTCNT register must be decremented to ‘`0`’. See **Section 7.2 “Receiving Packets”** for more information about clearing the EPKTCNT register. If the last data packet in the receive buffer is processed, EPKTCNT will become zero and the PKTIF bit will automatically be cleared.

12.2 Wake-On-LAN/Remote Wake-up

Wake-On-LAN or Remote Wake-up is useful in conserving system power. The host controller and other subsystems can be put in Low-Power mode and be woken up by the ENC28J60 when a wake-up packet is received from a remote station. The ENC28J60 must not be in Power-Save mode and the transmit and receive modules must be enabled in order to receive a wake-up packet. The ENC28J60 wakes up the host controller via the $\overline{\text{INT}}$ signal when the Interrupt Mask registers are properly configured. The receive filter can also be set up to only receive a specific wake-up packet (see Register 8-1 for available options). **Section 12.2.1 "Setup Steps for Waking Up on a Magic Packet"** shows the steps necessary in configuring the ENC28J60 to send an interrupt signal to the host controller upon the reception of a Magic Packet.

12.2.1 SETUP STEPS FOR WAKING UP ON A MAGIC PACKET

1. Set ERXFCON.CRCEN and ERXFCON.MPEN.
2. Service all pending packets.
3. Set EIE.PKTIE and EIE.INTIE.
4. Set up the host controller to wake-up on an external interrupt $\overline{\text{INT}}$ signal.
5. Put the host controller and other subsystems to Sleep to save power.

Once a Magic Packet is received, the EPKTCNT is incremented to '1', which causes the EIR.PKTIF bit to set. In turn, the ESTAT.INT bit is set and the $\overline{\text{INT}}$ signal is driven low, causing the host to wake-up.

